Learn to Hack Hackathon

Group 1 – Matrix Not Found

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Problem Statement

Buildings are a major contributor to carbon emissions. We built an **ML model** that **predicts a building's energy use intensity** (site EUI) from known factors, such as:

- Floor area and year built
- Energy Star rating
- Facility type and location
- Climate: temperatures, degree days, precipitation and snow

Buildings that use far more energy than the model expects for similar buildings are the ones leaders should **retrofit first**.

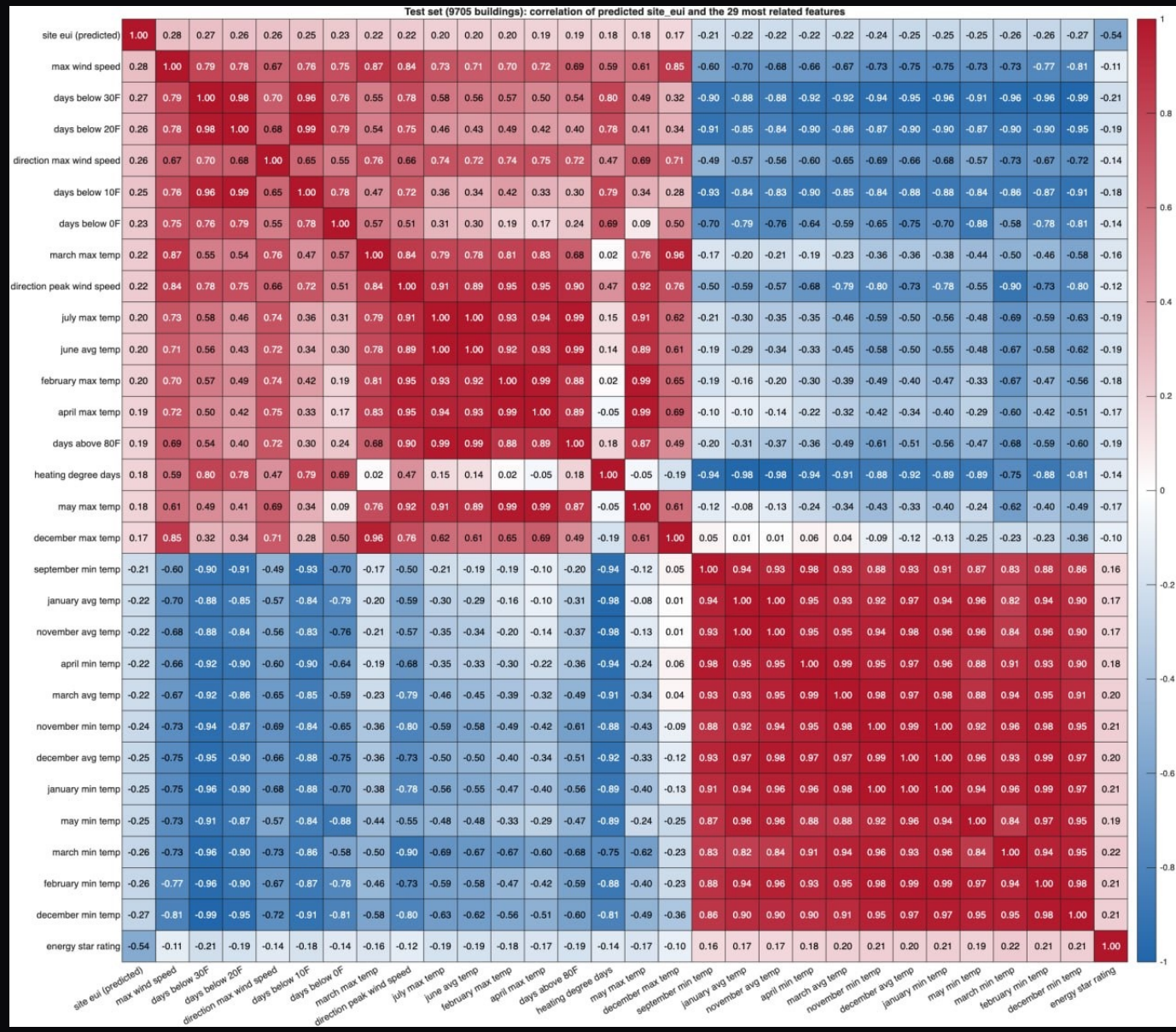


Exploratory Data Analysis

75,757 training building-years; **9,705** test buildings.

- The test set is one **unseen year** (year 7) with no State 6 buildings, so we validate by holding out a whole year.
- Energy Star rating is 35% missing: we filled gaps with **training-only medians** and added "was missing" flags (no rows dropped).

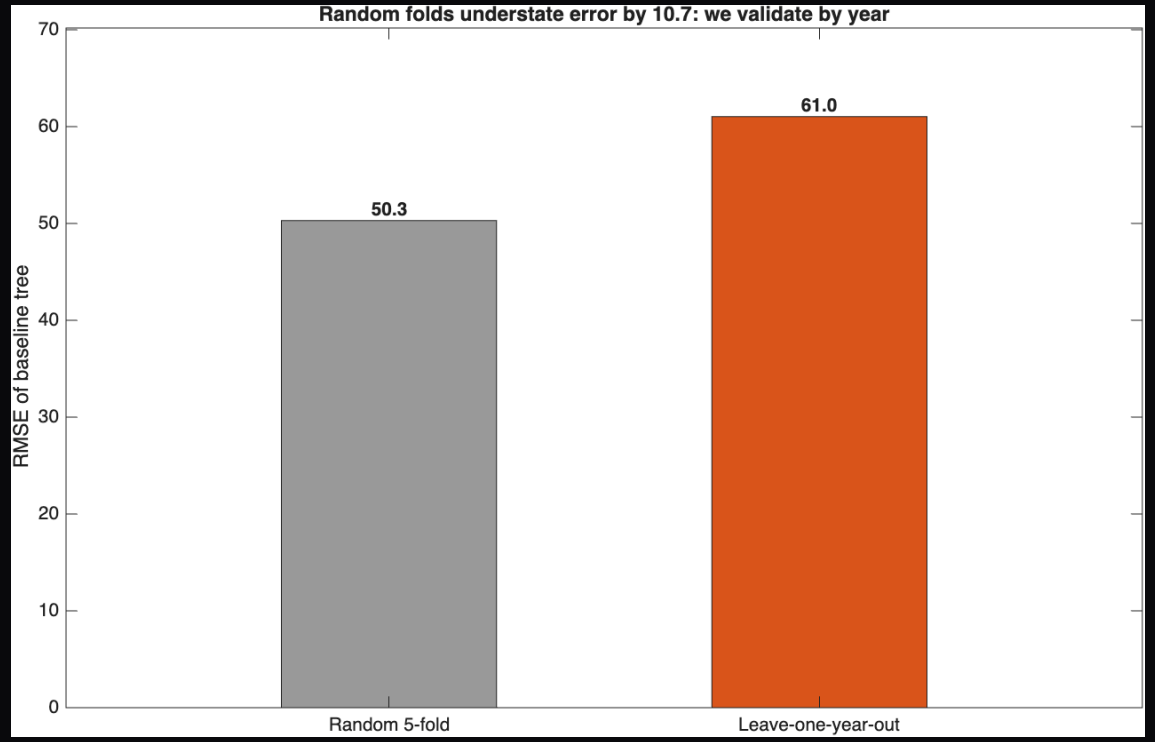
On the test set, **Energy Star rating** has the strongest correlation with site EUI (**−0.54**). Climate variables form tightly correlated blocks.



Model Architecture

Gradient-boosted regression trees (MATLAB **LSBoost**, the native XGBoost-family method): mixed numeric and categorical data, no scaling.

- **Tree** baseline → **LSBoost v1** (all features) → **v2** (shrinkage) → **v3** (feature selection: drop 36 monthly temperatures)
- Engineered features: building age, floor area × heating/cooling degree days, Energy Star × floor area
- **Leave-one-year-out** validation, with cleaning refit inside every fold (no leakage)



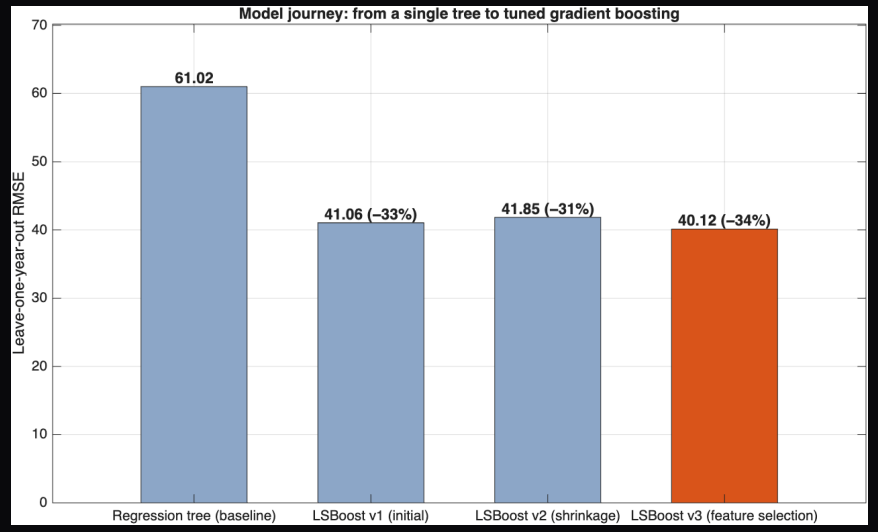
Results

Leave-one-year-out (official model)

LSBoost v3 is our final model:

- LOYO RMSE **40.1**, **34%** lower than a single tree
- MAE **21.0** kBtu/ft²/yr and R^2 **0.53** on years it never saw
- Best in **5 of 6** held-out years
- Reproducible: a clean rerun gives a byte-identical submission

Model	All years	Year 6	vs. tree
Tree	61.02	59.37	–
LSBoost v1	41.06	41.08	−33%
LSBoost v2	41.85	42.57	−31%
LSBoost v3	40.12	41.31	−34%



Model Comparison

Controlled experiment: same data and folds, only the model changes.

- **M1**: tree, all cleaned factors
- **M2**: tree, 8 selected factors
- **M3**: LSBoost, 8 selected factors

Boosting on selected features wins on **every metric** under both validations, consistent with the official result.

Five-fold pooled out-of-fold

Metric	M1	M2	M3
MAE	23.3941	23.3385	21.0390
MSE	2044.3723	2024.7375	1695.0047
RMSE	45.2147	44.9971	41.1704
R^2	0.3976	0.4034	0.5005

Latest-year holdout (train years 1–5, predict 6)

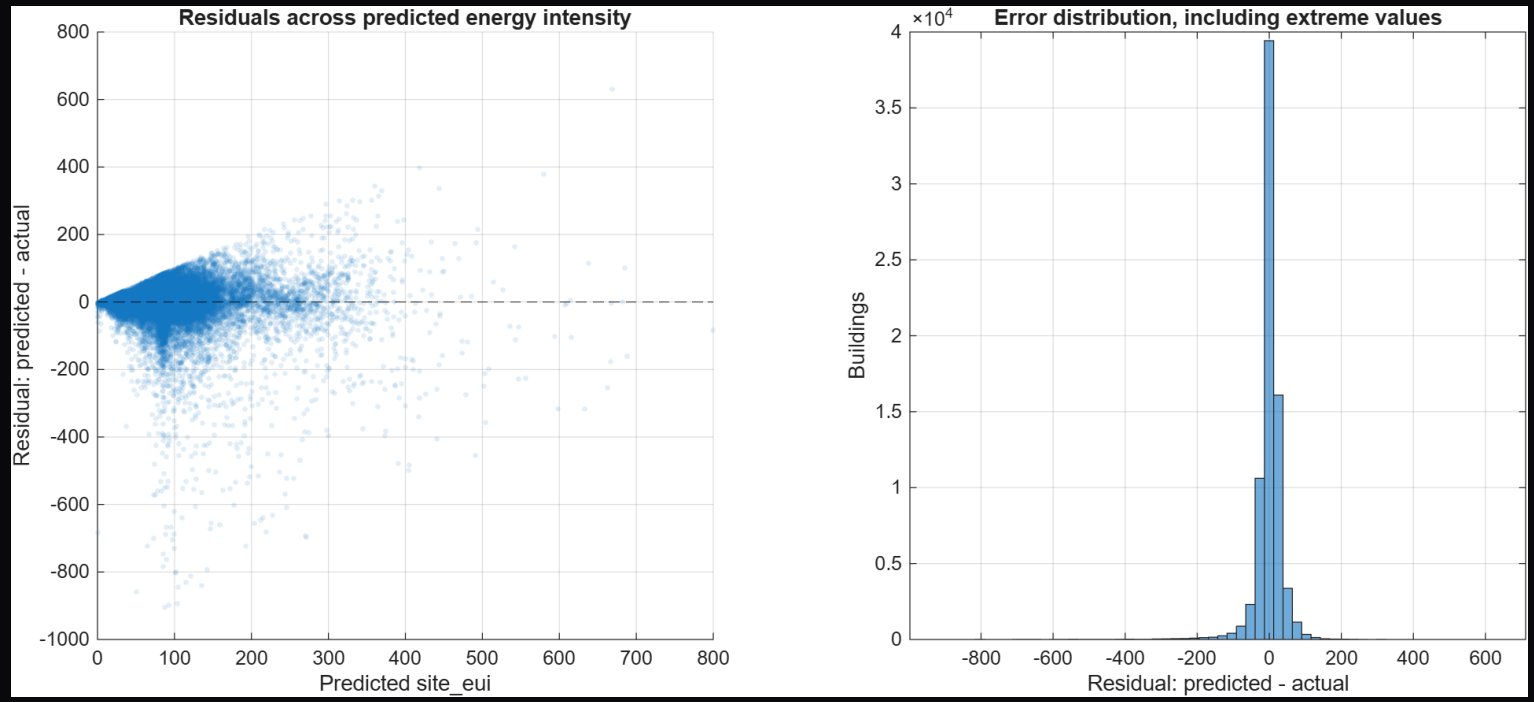
Metric	M1	M2	M3
MAE	22.7927	22.7324	21.0440
MSE	2023.2171	2003.8279	1842.3056
RMSE	44.9802	44.7641	42.9221
R^2	0.3777	0.3836	0.4333

Results - Residual

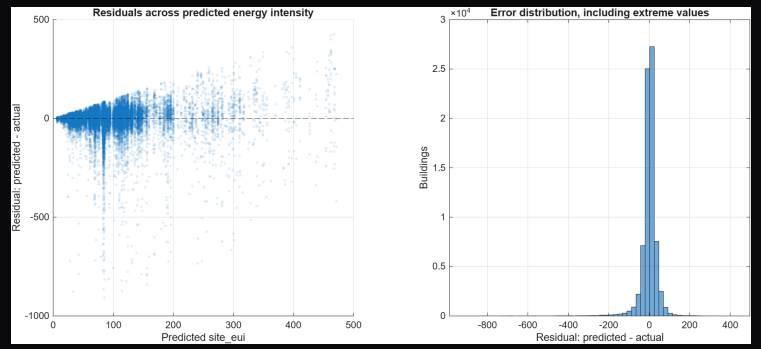
Residuals (predicted − actual) for all three models, five-fold out-of-fold:

- Centred on zero (M3 mean bias **0.03**)
- M3 has the tightest spread
- Long negative tail: a few buildings use **far more** than predicted

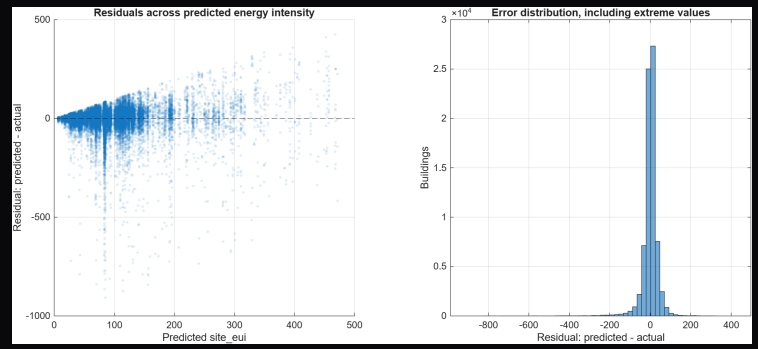
Those outliers are exactly what **Retrofit Radar** flags.



M3: LSBoost, selected factors (best)



M1: tree, all factors



M2: tree, selected factors

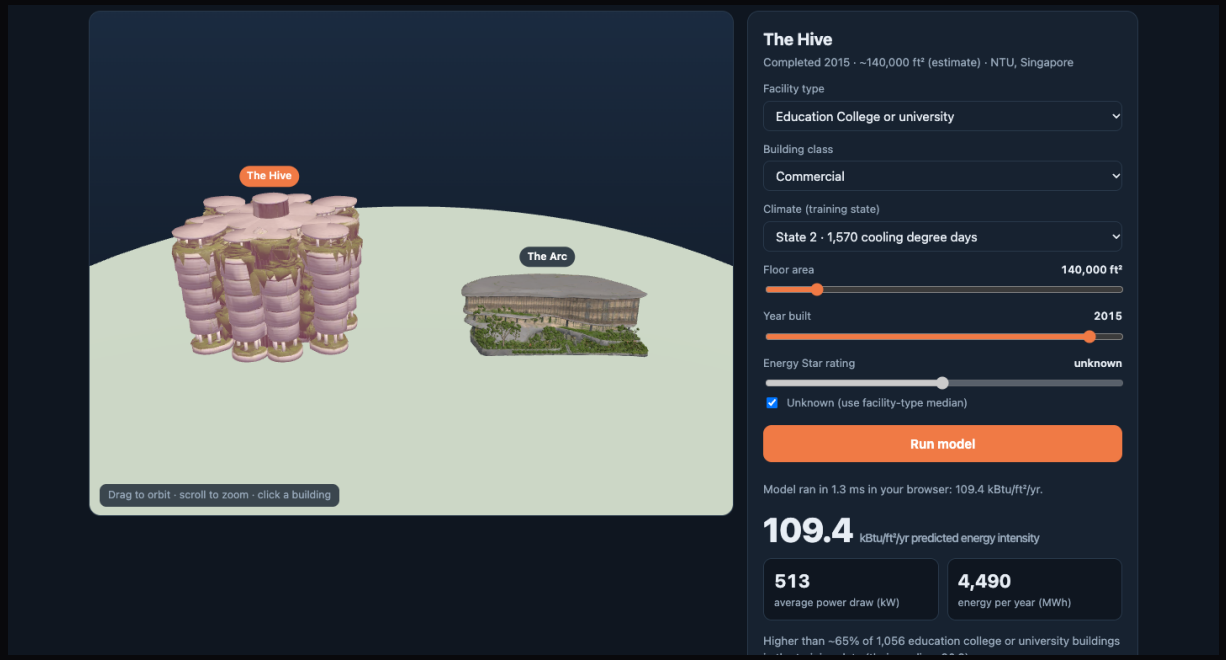
Live Demo

Retrofit Radar compares each building's actual energy use with what the model expects for similar buildings:

- **1,316** buildings use 50%+ more than expected
- The top 50 account for **1,573 GWh/yr** of excess energy

3D simulator: click NTU's The Hive or The Arc and our trained model runs **in the browser**, matching MATLAB exactly.

Open **index.html** in the repository for the dashboard, simulator and report.



Conclusion

Our best model is **LSBoost v3 (feature selection)**: LOYO RMSE **40.1**, MAE **21.0**, R^2 **0.53**, with 34% lower error than a single tree.

In real-world use, **Retrofit Radar** turns predictions into a ranked audit list: the buildings using far more energy than their peers.

Repository link:

[Roh00t/MatrixNotFound: LearnToHack]

